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1. What CropLabel does

CropLabel is a browser-based tool for labeling **crop plants, weeds and soil** in field plot images (UAV tiles, plot photos). It produces both:

- **Semantic segmentation** — every pixel classed as soil, crop or weed;
- **Instance segmentation** — every individual plant separated with its own ID, class, score and bounding box.

Each image contains **one crop species** (barley, wheat, canola, oats, ..., selected per image) plus possible weeds. Soil is labeled **completely automatically** — you never touch it.

Semantic value	Class	Color
0	unlabeled (vegetation not yet assigned)	—
1	soil (automatic — includes stubble/straw)	brown
2	crop	green
3	weed	red

2. The idea: label two plants, get the whole folder

The tool is built around a feedback loop between you and two AI systems:

1. **Soil is free.** Opening an image instantly separates living vegetation from everything else (Excess-Green index), so soil, stubble and plot markers are already labeled.
2. **One click labels all similar plants.** Click a single example plant with the ★ tool — SAM3.1 finds every similar-looking plant in the image and labels each one as a separate instance. Do the same once for a weed with the Weed class selected.
3. **Your corrections train a classifier.** Every time you press Save, a crop-vs-weed classifier retrain on all your labeled instances (color, shape and texture features).
4. **The folder labels itself.** " ⚡ Label ALL remaining" then runs on every image: automatic soil + SAM3.1 plant detection + the classifier deciding crop-vs-weed per plant. Review the results, fix the few mistakes, save — each save makes the next batch better.

Measured on barley trial tiles (1600×1600): ~40 plant instances found per image in under half a second on the GPU; a 38-image folder batch-labeled in about 2 minutes.

3. Requirements & installation

- Windows 10/11 64-bit, **NVIDIA GPU** (8 GB+ VRAM recommended) with a current driver
- Python 3.13 64-bit
- A modern browser. The app runs locally at `http://127.0.0.1:8323` — no internet needed after installation, no image ever leaves the computer.

1. Clone the repository and install dependencies:

```
git clone https://github.com/appndpird/CropLabel.git
cd CropLabel
pip install fastapi uvicorn opencv-python numpy pillow tokenizers
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128
pip install git+https://github.com/facebookresearch/sam3.git
pip install einops triton-windows pycocotools timm
pip install "numpy>=2.1" --upgrade # sam3 pins numpy<2 - undo that
```

2. Download the SAM3.1 checkpoint `sam3.1_multiplex.pt` from `huggingface.co/strangervisionhf/sam3.1-st-bf16` (open access, no account needed) and place it in the repo's `models\` folder.
3. Start with `CropLabel.bat` (or `python run_app.py`). The browser opens automatically; keep the console window open while working.

Important: torchvision must be a CUDA build (`+cu128`). A CPU-only torchvision crashes the SAM3.1 engine with a `roi_align` error — reinstall with the `--index-url .../cu128` shown above.

4. First-time setup

1. Click  **Setup** in the header.
2. **Dataset folder** —  Browse to the folder holding the images. Images are found *recursively*, so plot subfolders work (e.g. `tiles\plot_10001\...\png`). Supported: png, jpg, tif, bmp.
3. **Output folder** — where the label files are written (the input folder structure is mirrored there).
4. Leave the other settings at their defaults to start (section 11 explains each one). Press **Save settings** — the image list appears in the left sidebar, grouped by plot/folder.

5. The labeling workflow, step by step

1. **Pick an image** in the sidebar. Dots: grey = pending, blue = auto-labeled (review), green = reviewed, orange = skipped. The image opens with **soil already labeled**.
2. **Set the crop type** once per image (dropdown in the toolbar: barley / wheat / canola / oats / other). It is stored with the labels.
3. **Label the crop**: with the **Crop** class selected and the ★ tool active, click one clear example plant. All similar plants get labeled as individual instances (white outlines show the separation). Add another example click if some were missed; **Shift** + click anything wrongly labeled as a negative example — the search re-runs refined.
4. **Label weeds** (if any): select the **Weed** class, ★ click one weed. Or use the 🗂 Find box: type what the weed looks like (e.g. *broad leaf plant*) and press Enter.
5. **Review what the auto-label produced**. Tick **Instance numbers** and/or **Bounding boxes** in the option bar to see every plant's id and extent; tick **Side by side** to see the original image next to the annotated one. Move around with the mouse wheel (zoom), 🖱 hand tool / right-drag / **Space** + drag (pan), **F** (fit).
6. **Correct the mistakes** with the Fix tools (section 6b): **select** an instance (↗ or **Alt** + click), then ✍ **brush missing pixels into it**, ⌘ **split** it when two plants were merged (drag a line across, or click one seed per plant), ⌘ merge, 🗑 delete, 🔄 reclassify. Draw a missed plant with the 📐 **polygon** tool (click its corners). **Ctrl** + **Z** undoes anything (25 steps).
7. 📁 **Save** (or **Ctrl** + **S**) — writes all label files (plus the bounding-box files if **Save bbox files** is ticked), records the **summary statistics** (instance counts, plants / m², greenness ratio ..., section 10b) and retrains the crop-vs-weed classifier with your instances. Switching image auto-saves, so nothing is lost.
8. After 2–3 reviewed images, press ⚡ **Label ALL remaining** and let the folder label itself, then review the blue images.

6. All tools reference

The toolbar is grouped: **SAM** (automatic), **Create** (draw new labels by hand), **Fix** (correct existing instances) and **View** (move / zoom). Every tool shows a one-line hint in the top-left corner of the canvas when active.

6a. SAM & Create tools

Tool	Key	What it does
	1	Find similar — click ONE example plant (or drag a box around it): SAM3.1 labels every similar plant in the image as separate instances of the selected class. <code>Shift</code> + click = negative example (something it must NOT label). Examples accumulate and refine each other;  X clears them.
	2	Add one — labels just the single plant under the click as one instance (no search).
	3	Polygon — click the corners of a plant, then double-click or press <code>Enter</code> : the polygon becomes a new instance of the selected class. <code>Backspace</code> removes the last corner, <code>Esc</code> cancels. With Snap polygon to vegetation ticked (default) only the green pixels inside the polygon are kept, so a rough polygon gives an exact plant outline. If an instance is <i>selected</i> , the polygon is added to that instance instead.
	4	Brush — with an instance selected it paints the missing pixels into that instance (the fix for partially labeled plants). Without a selection it paints the class only (semantic). Hold <code>Ctrl</code> while painting to also take pixels away from neighbouring instances. Size: slider, or <code>[</code> / <code>]</code> .
	5	Eraser — removes pixels from the selected instance only (precise clean-up). With nothing selected it erases any label back to soil.

6b. Fix tools — correcting the auto-labels

Tool	Key	What it does
	6	Select — click an instance: it gets a yellow outline and a small panel (→ crop / → weed / 🗑 delete / ✕). While selected: 🖌 paints into it, ✖ erases from it, ⬢ adds to it, <code>Delete</code> removes it, <code>c</code> / <code>w</code> reclassify it, the ✂ tools split it. <code>Alt</code> +click selects with any tool; <code>Esc</code> deselects.
	7	Split by line — drag a line right across a merged plant (start and end outside it). The instance is cut along the line into separate instances; no gap is left.
	8	Split by seeds — click once inside each plant of a merged instance (two or more seeds), then <code>Enter</code> or double-click. A watershed on the shape's distance transform splits it into one instance per seed, following the natural "neck" between the plants. Fastest way to separate 3–4 seedlings that SAM joined.
	9	Split by box — drag a box over the part of an instance that should be its own plant.
	P	Split by polygon — when several plants were labeled as one instance, click the corners around one of them exactly like the ⬢ polygon tool, then <code>Enter</code> , double-click or click the first corner. The part inside the polygon becomes its own instance; the rest keeps its number. Only pixels of that instance are affected, so the polygon can be rough.
	A	Auto split — click an over-merged instance: it is split automatically at the plant crowns (the wide points of the shape). <code>Shift</code> +click asks how many plants it holds and forces exactly that many parts. 🌟 Auto-label already applies this to every instance larger than Auto-split × the typical single-plant area (Setup, default 1.7).
	0	Merge — click the first instance, then the second: they become one instance (fixes one plant split into several).
	D	Delete — removes the instance under the click (pixels return to soil/unlabeled).
	R	Reclassify — the clicked instance becomes the selected class (crop ↔ weed).

Instance numbers are always sequential (1 ... N in the order the plants were created). When instance 13 is split into three, the parts become 13, 14 and 15 and every later instance moves up by two; a delete or merge closes the gap. The numbers drawn on the image, the ids in `_instances.png` and the JSON are always the same, so the files never have gaps. Bounding boxes and areas are recomputed after every edit.

Version 1.2 — fewer merged seedlings. When SAM3.1 returns both a group of touching seedlings and the single plants inside it, the single plants are now preferred (the old rule kept whichever scored higher, usually the group). The automatic split then fires at $1.7 \times$ the typical single-plant area instead of $2.5 \times$ — two merged 2-leaf seedlings are $\sim 1.7\text{--}2.3 \times$, a single 3-leaf plant $\sim 1.3\text{--}1.6 \times$. On the 40 OzBarley sample tiles this cut the number of over-merged instances from 304 to 99. If a large 3–4-leaf plant is occasionally cut at a leaf,  merge fixes it with two clicks.

6c. View options and navigation

Option / control	Effect
Instance numbers	Draws each instance's id on top of the plant (also in the saved <code>_overlay.png</code>).
Bounding boxes	Draws a class-coloured bounding box around every instance (also in the saved overlay).
Save bbox files	On Save also writes <code><name>_boxes.json</code> , <code><name>_boxes.txt</code> (YOLO) and <code><name>_boxes.png</code> — see section 10.
Side by side	Shows the original image on the left and the annotated image on the right. Each panel moves and zooms on its own: the wheel zooms the panel under the cursor, dragging inside the left panel moves the original (any button), the hand / right-drag / Space-drag move the annotated panel. So you can zoom into one plant on the right and inspect the raw pixels of any other area on the left at the same time. Tools act on the right panel only.
 Link views	Locks the two panels to the same zoom and position (moving one moves the other) — handy to compare exactly the same spot. Untick to move them independently again.
Snap polygon to vegetation	Polygon tool keeps only vegetation pixels inside the drawn polygon.
 hand () , $-/+$, $\swarrow$ Fit, 1:1	Left-drag moves the image with the hand tool; right-drag, middle-drag or  +drag move it with any tool. Mouse wheel zooms at the cursor, arrow keys nudge,  fits the image to the window.
Overlay slider	Fades the label view to check the underlying image.

Other controls:  /  = select Crop/Weed class ·  +  /  undo/redo ·  +  save · 

Auto-label runs the full automatic pipeline on the current image · **Skip** marks an image to ignore · 

Summary opens the statistics panel.

7. The 🤖 "Find" box — telling the labeler what to look for

The text box above the canvas sends any free-text description to SAM3.1 as a concept prompt and labels the matches as the **currently selected class**:

- Select **Weed**, type *broad leaf plant* → broadleaf weeds get labeled.
- Select **Crop**, type *thin grass seedling* → grass-like seedlings get labeled.

Useful phrasings are visual descriptions: *plant, seedling, green plant, grass, leaf, small plant*. Species names (e.g. "ryegrass") usually work worse than descriptions of shape and color. If a prompt returns too much or too little, adjust the **Threshold** in Setup (lower = more detections) and run it again — every run is undoable.

SAM3.1 finds *plants* extremely well but cannot reliably tell crop from weed by name alone — both are green plants. That judgement comes from your examples: the ★ clicks and the classifier (next section).

8. The crop-vs-weed classifier

Every saved image contributes its instances (with your corrected classes) as training data. Once at least **3 crop and 3 weed instances** exist, the classifier activates (header pill turns green) and from then on every auto-label and batch run classifies each detected plant as crop or weed automatically, using color, shape and texture features.

- Until then, all detected plants default to **crop** (the safe assumption in a crop trial) — reclassify the weeds you see with 🗑️ and save.
- The classifier improves with every save; there is nothing to configure.
- The header pill shows its state, e.g. `classifier: 74 instances`.

9. Batch auto-labeling a whole folder

⚡ **Label ALL remaining** processes every pending image in the background: automatic soil → SAM3.1 plant instances (using the plant prompts from Setup) → crop/weed classification. Progress shows in the sidebar; you can keep labeling while it runs, and Stop it anytime.

- Results are saved with status **auto** (blue dot) — every one stays fully editable. Open, correct, Save to promote to reviewed (green).
- Speed: roughly 1–3 seconds per image on a GPU.
- Re-running the batch only touches images that are still pending.

Best practice: review 3–5 images first (including at least a few weed examples if the field has weeds), then batch. The classifier then labels crop vs weed for the whole folder instead of defaulting to crop.

10. Output files

Written to the output folder, mirroring the dataset's folder structure. **The semantic mask has the same filename and pixel size as the input image**, so downstream training pipelines can pair image ↔ label by name.

File	Content
<code><name>.png</code>	Semantic index mask (0/1/2/3), same name & size as the input image.
<code><name>_instances.png</code>	16-bit instance-ID map (0 = no instance) — the instance segmentation. IDs match the JSON records.
<code><name>_color.png</code>	Color-coded semantic mask.
<code><name>_overlay.png</code>	Labels blended over the image for quick visual checks.
<code><name>.json</code>	Crop type, status, engine, the summary statistics (section 10b), and one record per instance: id, class, score, bounding box, centroid, area (all at original resolution).
<code><name>_boxes.json</code>	<i>Optional</i> (Save bbox files): every instance's bounding box as <code>[x0, y0, x1, y1]</code> pixels with id, class, label, area and score.
<code><name>_boxes.txt</code>	<i>Optional</i> : the same boxes in YOLO format — one line per instance <code>class cx cy w h</code> (normalized 0–1), class 0 = crop, 1 = weed. Ready for detector training.
<code><name>_boxes.png</code>	<i>Optional</i> : the image with numbered boxes drawn, for checking.
<code>records.json</code>	Queue status store (drives the sidebar colors) — also holds every image's summary statistics.
<code>summary.csv</code>	One row per labeled image with all summary statistics — open directly in Excel.

10b. Summary statistics

Every Save computes these for the image and writes them to `<name>.json`, `records.json` and `summary.csv`. The  **Summary** button shows the current image and the averages/totals over the whole dataset.

Statistic	Meaning
n_instances , n_crop_instances , n_weed_instances	Number of labeled plants, by class.
plants_per_m2 , weeds_per_m2	Crop / weed instances divided by the image ground area. Needs the GSD (mm / pixel) of the input images, entered in ⚙ Setup; blank until then.
image_area_m2 , crop_cover_m2	Ground area of the image and of the crop pixels (needs GSD).
crop_pct , weed_pct , soil_pct , unlabeled_pct	Percentage of image pixels in each class. <i>unlabeled</i> = vegetation not assigned to any instance — a quick completeness check.
greenness_ratio	Fraction of image pixels that are living vegetation (Excess-Green index above the ExG threshold) — the canopy-cover / greenness measure.
mean_exg , mean_exg_crop	Mean Excess-Green index over the whole image and over the crop pixels only (plant vigour proxy).
labeled_veg_fraction	Share of the vegetation pixels that belong to a labeled instance (1.0 = everything green is labeled).
mean_crop_area_px , median_crop_area_px , mean_crop_area_cm2	Plant size statistics (cm ² needs GSD).

11. Settings reference (⚙️ Setup)

Setting	Default	Meaning
Plant prompts	plant, seedling	SAM3.1 text prompts used by 🌟 auto-label and the batch. Comma-separated; results are merged and de-duplicated.
Threshold	0.35	SAM3.1 confidence cut-off. Lower finds more (and more false) plants. 0.35 gives more single-plant detections for small seedlings.
ExG	0.06	Excess-Green vegetation threshold. Lower catches paler/yellowish plants as vegetation; higher pushes more into soil. Also clips every SAM instance so masks never bleed into soil.
Min px	30	Instances smaller than this many pixels (at working resolution) are dropped as noise.
Auto-split ×	1.7	After 🌟 auto-label, instances larger than this × the typical single-plant area are split automatically at the plant crowns. 0 = off. For 2-leaf seedlings keep 1.7; if most plants have 4+ leaves and you see single plants being cut, raise to 2.0–2.5.
GSD (mm / pixel)	empty	Ground sampling distance of the input images. Enables plants / m ² , weeds / m ² , crop cover m ² and plant area cm ² in the statistics. For a tile cut from an orthomosaic use the orthomosaic's resolution (e.g. 5 mm/px → 5).
Instance numbers / Bounding boxes / Save bbox files / Side by side / Snap polygon	off / off / off / off / on	The option-bar checkboxes; remembered between sessions (settings.json).
work_res	1024	Internal working resolution (settings.json). Labels are always saved at the original image size.

12. Troubleshooting

Problem	Fix
Header pill: <i>SAM3.1 NOT available</i>	No NVIDIA GPU/driver, or <code>models\sam3.1_multiplex.pt</code> missing, or torchvision is a CPU build (reinstall with <code>--index-url .../cu128</code>). Hover the pill for the exact error.
★ finds only a few plants	Click a <i>typical, clearly visible</i> plant; add a second example click. Dragging a snug box around one plant often beats a click. Lower the Threshold for more matches.
Pale/yellow plants counted as soil	Lower the ExG setting (e.g. 0.03) and reopen the image.
Stubble/straw labeled as plants	Raise ExG slightly, raise Min px, or <code>Shift</code> +click a piece of stubble as a ★ negative example.
Everything labeled crop, weeds missed	Expected before the classifier has weed examples. Reclassify () a few weeds, Save, re-run auto-label / batch.
Two or more touching plants merged into one instance	Use  split by seeds : click once inside each plant, press <code>Enter</code> . Or drag a  line across the join. (Undo if the cut is not where you wanted and try again.)
One plant split into several instances	 merge : click one part, then the other.
Part of a plant is not labeled	Select the instance ( or <code>Alt</code> +click), then paint the missing part with  — it joins that instance. Or  polygon around the missing part while the instance is selected.
A plant was missed completely	 click it, or draw it with  polygon (corners → Enter).
plants / m ² is empty in the summary	Enter the GSD (mm / pixel) in  Setup, Save settings, then Save the image again.
App won't start / port in use	Only one instance at a time — port 8323. Close the other console window.
Wrong folder scanned	 Setup → check both folders → Save settings. The sidebar reloads immediately.